

Experiment No:2

Title : Implementation of DML commands of SQL with suitable examples

- Insert table
- Update table
- Delete Table

Objective :

To understand the different issues involved in the design and implementation of a database system

To understand and use data manipulation language to query, update, and manage a database

Theory :

DATA MANIPULATION LANGUAGE (DML): The Data Manipulation Language (DML) is used to retrieve, insert and modify database information. These commands will be used by all database users during the routine operation of the database. Let's take a brief look at the basic DML commands:

1. INSERT

2. UPDATE

3. DELETE

1. INSERT INTO: This is used to add records into a relation. These are three type of INSERT INTO queries which are as

a) Inserting a single record

Syntax: INSERT INTO < relation/table name> (field_1,field_2.....field_n)VALUES
(data_1,data_2, data_n);

Example: SQL>INSERT INTO student(sno,sname,class,address)VALUES
(1,'Ravi','M.Tech','Palakol');

b) Inserting a single record

Syntax: INSERT INTO < relation/table name>VALUES (data_1,data_2,data_n);

Example: SQL>INSERT INTO student VALUES (1,'Ravi','M.Tech','Palakol');

c) Inserting all records from another relation

Syntax: INSERT INTO relation_name_1 SELECT Field_1,field_2,field_n
FROM relation_name_2 WHERE field_x=data;

Example: SQL>INSERT INTO std SELECT sno,sname FROM student
WHERE name = 'Ramu';

d) Inserting multiple records

Syntax: INSERT INTO relation_name field_1,field_2,...field_n) VALUES
(&data_1,&data_2,.....&data_n);

Example: SQL>INSERT INTO student (sno, sname, class,address)
VALUES (&sno,'&sname','&class','&address');

Enter value for sno: 101
Enter value for name: Ravi
Enter value for class: M.Tech
Enter value for name: Palakol

2. UPDATE-SET-WHERE: This is used to update the content of a record in a relation.

Syntax: SQL>UPDATE relation name SET Field_name1=data,field_name2=data,
WHERE field_name=data;

Example: SQL>UPDATE student SET sname = 'kumar' WHERE sno=1;

3. DELETE-FROM: This is used to delete all the records of a relation but it will retain the structure of that relation.

a) DELETE-FROM: This is used to delete all the records of relation.

Syntax: SQL>DELETE FROM relation_name;

Example: SQL>DELETE FROM std;

b) DELETE -FROM-WHERE: This is used to delete a selected record from a relation.

Syntax: SQL>DELETE FROM relation_name WHERE condition;

Example: SQL>DELETE FROM student WHERE sno = 2;

5. TRUNCATE: This command will remove the data permanently. But structure will not be removed.

Difference between Truncate & Delete:-

- ✓ By using truncate command data will be removed permanently & will not get back where as by using delete command data will be removed temporally & get back by using roll back command.
- ✓ By using delete command data will be removed based on the condition where as by using truncate command there is no condition.
- ✓ Truncate is a DDL command & delete is a DML command.

Syntax: TRUNCATE TABLE <Table name>

Example TRUNCATE TABLE student;

- **To Retrieve data from one or more tables.**

1. SELECT FROM: To display all fields for all records.

Syntax : SELECT * FROM relation_name;

Example : SQL> select * from dept;

DEPTNO	DNAME	LOC
-----	-----	-----
10	ACCOUNTING	NEW YORK
20	RESEARCH	DALLAS
30	SALES	CHICAGO
40	OPERATIONS	BOSTON

2. SELECT FROM: To display a set of fields for all records of relation.

Syntax: SELECT a set of fields FROM relation_name;

Example: SQL> select deptno, dname from dept;

DEPTNO	DNAME
-----	-----
10	ACCOUNTING
20	RESEARCH
30	SALES

3. SELECT - FROM -WHERE: This query is used to display a selected set of fields for a selected set of records of a relation.

Syntax: SELECT a set of fields FROM relation_name WHERE condition;

Example: SQL> select * FROM dept WHERE deptno<=20;

DEPTNO	DNAME	LOC
-----	-----	-----
10	ACCOUNTING	NEW YORK
20	RESEARCH	DALLAS

LAB PRACTICE ASSIGNMENT:

Create a table EMPLOYEE with following schema:

(Emp_no, E_name, E_address, E_ph_no, Dept_no, Dept_name, Job_id , Salary)

Write SQL queries for following question:

1. Insert atleast 5 rows in the table.
2. Display all the information of EMP table.
3. Display the record of each employee who works in department D10.
4. Update the city of Emp_no-12 with current city as Nagpur.
5. Display the details of Employee who works in department MECH.
6. Delete the email_id of employee James.
7. Display the complete record of employees working in SALES Department.